Say my [*insert name*]: Participant lexical selection increases participant engagement

Haley Hsu[1] & Brandon Papineau[2], <haleyhsu@usc.edu>,
[1]University of Southern California & [2]Stanford University

Experimentally elicited data in the study of human behavior is fundamentally contingent upon the collection of *reliable* data. Although crowdsourcing platforms (e.g., Prolific, MTurk, etc.) have made data collection more efficient and accessible, there continue to be reported issues of participant disengagement (e.g. [1,2]) for both online and in-lab experimental data collection. Depending on measurement instrument and threshold strictness, prior work in linguistics has estimated that 3-73% of participants provide careless responses or lack of careful attention results ([3,4]); regardless of exact amount, consensus remains that careless responses constitute a sizable proportion of experimental data ([5]). Given the often repetitive nature of reading or listening to trivial and/or impersonal stimuli in experimental settings for language studies, collected behavioral linguistic data may suffer from a *lack of participant engagement*, especially when such tasks are contrasted with the high-load tasks of everyday interactions.

To combat this lack of participant engagement, researchers often include attention checks and/or set accuracy thresholds, resulting in data exclusion after collection has concluded. While these strategies help prevent poor data quality, both result in the *removal of data* and do not resolve the lack of participant engagement. It has been proposed that increasing participant engagement in linguistic studies might elicit a more accurate profile of real-world language processing ([2]). As a first attempt to this, we propose an experimental feature that allows participants to insert their lexical selections (e.g., names) to be distributed throughout the experiment, *improving* the likelihood participants feel more invested in the task.

Hyp1: No effect? Shallow processing (e.g. [6]) is such a strong cognitive default that we process all information, regardless of if we have any agency in forming the information, similarly.

Hyp 2: Agency? The inclusion of chosen lexical items in an experiment helps modulate the participants' engagement and performance during the task, such that attention is higher.

Experiment. To test this, we conduct a rapid serial visual presentation (RSVP, 150ms rate) task, where U.S.-born native English speakers (target n ≈ 100) provide twelve names, read a sentence, then recall that sentence by filling in a missing word from the sentence (Fig. 1). We manipulate whether participants see preset names or their chosen names (see *ex.1*, 24 targets, 24 fillers, Latin-Square). Targets are all active, plausible sentences (adapted from [6]) and contain only one name, which serves as the critical blank region for the fill-in-the-blank task. Target sentences do not contain any gender-specific terminology. Critical regions for the fillers (adapted from [7,8]) appear in different locations. In order to avoid recency bias from their chosen names, participants were also given a list of all preset names at the start and were instructed not to repeat any names. Preset names shared similar frequency values ([10]).

Results. In order to probe levels of engagement, we assume better recall reflects more engagement to the task (e.g., [9]). Thus, we predict a higher accuracy score and faster reaction times for sentences containing names participants provide than sentences with preset names. Our results (Figs. 2,3) demonstrate this pattern (*p's* < 0.05), supporting our agency hypothesis.

In sum, our results indicate that participants are more engaged in a task when stimuli include their lexical selections. Increasing participant engagement may yield higher data quality for language studies, better reflect real-world language processing, and assure that financial resources are allocated to rewarding highly engaged research participants appropriately.

ex. 1) Experimental design *(fill-in-the-blank locations are counterbalanced for fillers, sentences were approximately all the same length)*

Preset name:	No one knew **Daisy** loved the puppy so much.
Chosen name:	No one knew [name] loved the puppy so much.
Trial task:	No one knew _____ loved the puppy so much.

Fig. 1. Sample item (*instructions and practice items explained task, fixation cross on screen for 1s before each sentence started, sentence was presented one word at a time at a rate of 150ms per word, sentence was followed by a fill-in-the-blank task immediately after*).

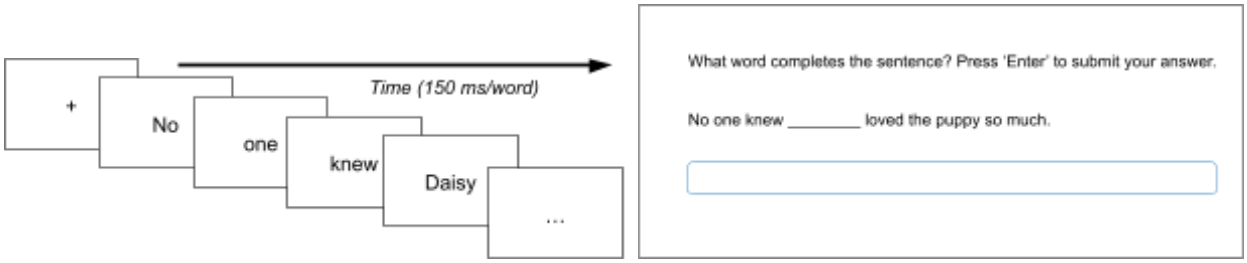


Fig. 2. Response times (ms) by condition.

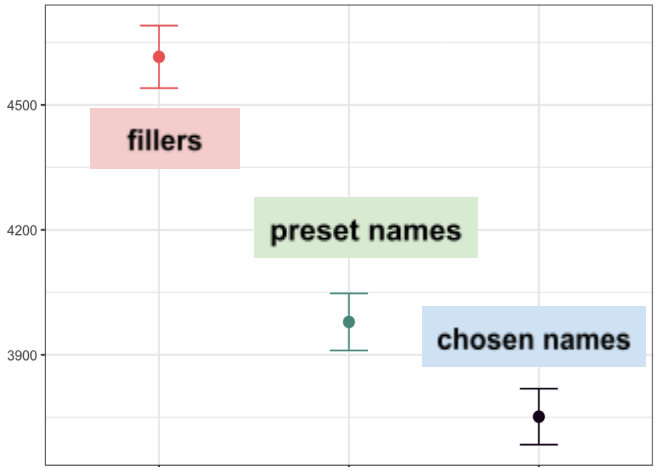
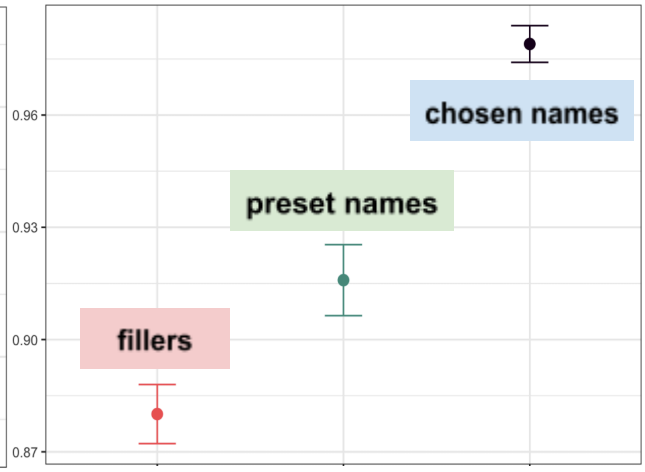


Fig. 3. Response accuracy by condition.



References [1] Eyal et al. (2021). Data quality of platforms and panels for online behavioral research. *Behav Res.* [2] Christianson et al. (2022). Chapter Two - What if they're just not that into you (or your experiment)? On motivation and psycholinguistics. *Psych of Learning and Motivation.* [3] Maniacci & Rogge (2014). Caring about carelessness: Participant inattention and its effects on research. *Journal of Research in Personality.* [4] Peer et al. (2017). Beyond the Turk: Alternative platforms for crowdsourcing behavioral research. *JESP.* [5] Brühlmann et al. (2020). The effectiveness of warning statements in reducing careless responding in crowdsourced online surveys. *Behav Res.* [6] Ferreira (2003). The misinterpretation of noncanonical sentences. *Cog Psych.* [7] Patson et al. (2009). Lingering misinterpretations in garden-path sentences: evidence from a paraphrasing task. *JEP.* [8] Fodor et al. (2017). Center-Embedded Sentences: What's Pronounceable Is Comprehensible. *On Concepts, Modules, and Language: Cognitive Science at Its Core.* [9] Song et al. (2021). Neural signatures of attentional engagement during narratives and its consequences for event memory. *PNAS.* [10] Social Security Administration (2025) [https://www.ssa.gov/oact/babynames/].